



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

INTERNAL HACKATHON 2026

INNOVATE • COLLABORATE • IMPACT

CIT CAMPUS | STUDENT INNOVATION | 2026

Project: CareFor — Multi-Modal AI Non-Invasive Health & Diagnostic Suite

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The Challenge

Define the problem clearly before presenting the technology.

PROBLEM STATEMENT

- **Limited Screening Access:** Frequent health monitoring is difficult where medical equipment and healthcare professionals are not readily available.
- **Untapped Smartphone Potential:** A smartphone can capture facial, eye, voice, cough, and movement signals, but these capabilities remain largely fragmented.
- **No Personal Health Baseline:** Existing solutions often measure isolated parameters rather than continuously identifying meaningful changes from an individual's normal health pattern.

TARGET USERS

- **Beneficiaries:** Rural & low-resource populations, elderly individuals, chronic care patients, and remote workers.
- **Stakeholders:** Family caregivers, village health workers, doctors, campus clinics, & telemedicine providers.
- **Context / Location:** Primary healthcare centers, home care, and remote diagnostic screening.

TECHNICAL & SOCIAL IMPACT

- **\$0 Extra Hardware Cost:** Turns any basic smartphone into a multi-modal health monitor without buying expensive medical devices.
- **Instant Universal Access:** Empowers rural communities & home patients to perform 1-minute non-invasive health checks anytime.
- **Early Risk Prevention:** Catches hidden warning signs in vitals, fatigue, stress, & breathing before emergencies occur.

PROBLEM IN ONE LINE

"How might we fuse smartphone camera video, eye tracking, speech acoustics, and cough sound waves into a single AI app for instant, zero-cost health screening?"

Existing System & Gap Analysis

Show what is already available and where the opportunity lies.

CURRENT SITUATION

• **Everyday Gadget Reliance:** Users buy multiple small gadgets (Smartwatch, Pulse Oximeter, Stethoscope, Thermometer).

• **Replaces Small Gadgets:** CareFor replaces small personal gadgets with 1 smartphone app (Not replacing big ICU machines).

PERSONAL MEDICAL DEVICES



EXISTING APPROACH

• **Single-Signal Limit:** Current commercial solutions bias only on 1 isolated scan (e.g. Watch pulse or Eye tracking alone).

• **Unmet Need:** Requires combining all 4 non-invasive bio-scans into one unified suite.

Apple Health
Face / Watch Pulse

tobii Tobii AI
Eye & Gaze Tracking

Empatica E4
Speech & Stress Bio

VitaloJAK
Cough Audio Monitor

UNMET NEED (CAREFOR GAP)

• **CareFor Advantage:** Zero-hardware multi-modal camera + microphone AI diagnostic suite.

• **Holistic AI Fusion:** Fuses 4 diagnostic scans into 1 app for instant baseline health tracking.



Proposed Solution

Present the idea in one glance: input → intelligence → action → outcome.



TRACK 01 // CAMERA

Face Scan

- **Camera Video:** Reads light absorption on facial skin in real time.
- **Blood Pulse Tracking:** Captures micro skin color changes each heartbeat.
- **Vital Signs Output:** Calculates Pulse Rate (BPM), SpO2 %, HRV & Respiration.

INPUT SENSOR
Smartphone Camera

CORE TECH
MediaPipe + rPPG + DSP

OUTPUT VITALS
Pulse Rate, SpO2 %, HRV,
Breaths

02

EYE SCAN

03

SPEECH SCAN

04

COUGH SCAN

04

Research Evidence

CareFor is built on existing research, not assumptions.

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MODALITY	RESEARCH SHOWS	OUR APPROACH
Face / rPPG	Facial video can capture pulse-related physiological changes.	POS/CHROM + Signal Processing
Eye Tracking	Eye movements and blinking can provide measurable behavioral and neurological indicators.	MediaPipe Iris + Temporal Analysis
Voice Analysis	Vocal characteristics can act as digital biomarkers for health-related conditions.	MFCC + Acoustic Features + ML
Cough Analysis	Cough acoustics contain patterns useful for respiratory screening.	Mel-Spectrogram + CNN

Core Technical Innovation

What makes CareFor uniquely breakthrough compared to traditional devices and single-signal apps.

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PILLAR 01 // ACCESSIBILITY

\$0 Extra Hardware

- **Standard Phone Sensors:** Uses built-in RGB camera & microphone only.

- **Replaces Small Gadgets:** Eliminates smartwatches, pulse oximeters, & digital stethoscopes.

- **Zero Cost Barrier:** 1-minute non-invasive screening accessible to anyone, anywhere.

PILLAR 02 // MULTI-MODAL AI

4-Scan AI Fusion

- **Full-Body Signal Fusion:** Fuses facial skin, eye gaze, speech acoustics, & cough audio.

- **Advanced AI Engines:** MediaPipe + rPPG + DSP + Audio Mel-Spectrogram CNN.

- **Holistic Baselines:** Tracks cardiac, respiratory, & neurological state together.

PILLAR 03 // ACTIONABLE CARE

Healio AI Assistant

- **Plain Health Guidance:** RAG medical chatbot translates numbers into easy advice.

- **Persistent Health Log:** Automatically tracks daily, weekly, & monthly vital trends.

- **Remote Guardian Alerts:** Notifies family & doctors if vital warning signs trigger.

INNOVATION IN ONE LINE

"CareFor replaces multiple separate health gadgets by fusing camera video, eye tracking, voice acoustics, and cough sound waves into a single zero-cost, 1-minute AI screening app."

How It Works

A simple end-to-end workflow: input → pre-process → intelligence → decision → output.

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01 // INPUT

Sensor Capture

- **Camera Video:**

Captures 30 FPS face video & upper-face eye close-up.

- **Mic Audio:**

5s voice sample + 2 cough sound bursts.

- **Zero Hardware:**

Standard phone camera & mic (\$0 cost).

02 // PRE-PROCESS

Signal Cleaning

- **MediaPipe:**

Locks facial landmarks & tracks iris boundaries in real time.

- **rPPG Filter:**

POS optical model + Butterworth filter wipes motion noise.

- **Audio VAD:**

Strips room echoes & ambient background noise.

03 // INTELLIGENCE

Multi-Track AI

- **rPPG Engine:**

Computes Pulse Rate, SpO2 %, HRV & Breaths/min.

- **PERCLOS Model:**

Measures eye closure ratio % & gaze speed.

- **Acoustic AI:**

Pitch trembles & audio CNN cough patterns.

04 // DECISION

Clinical Decision

- **Biomarker Fusion:**

Combines vitals, fatigue, stress & cough strain.

- **Risk Scoring:**

Compares results against baseline thresholds.

- **Early Alerts:**

Flags vital drops before emergencies occur.

05 // OUTPUT

Actionable Output

- **Vitals Dashboard:**

Real-time visual display of all 4 scan results.

- **Healio Chatbot:**

Explains technical numbers in plain language.

- **Medical Diary:**

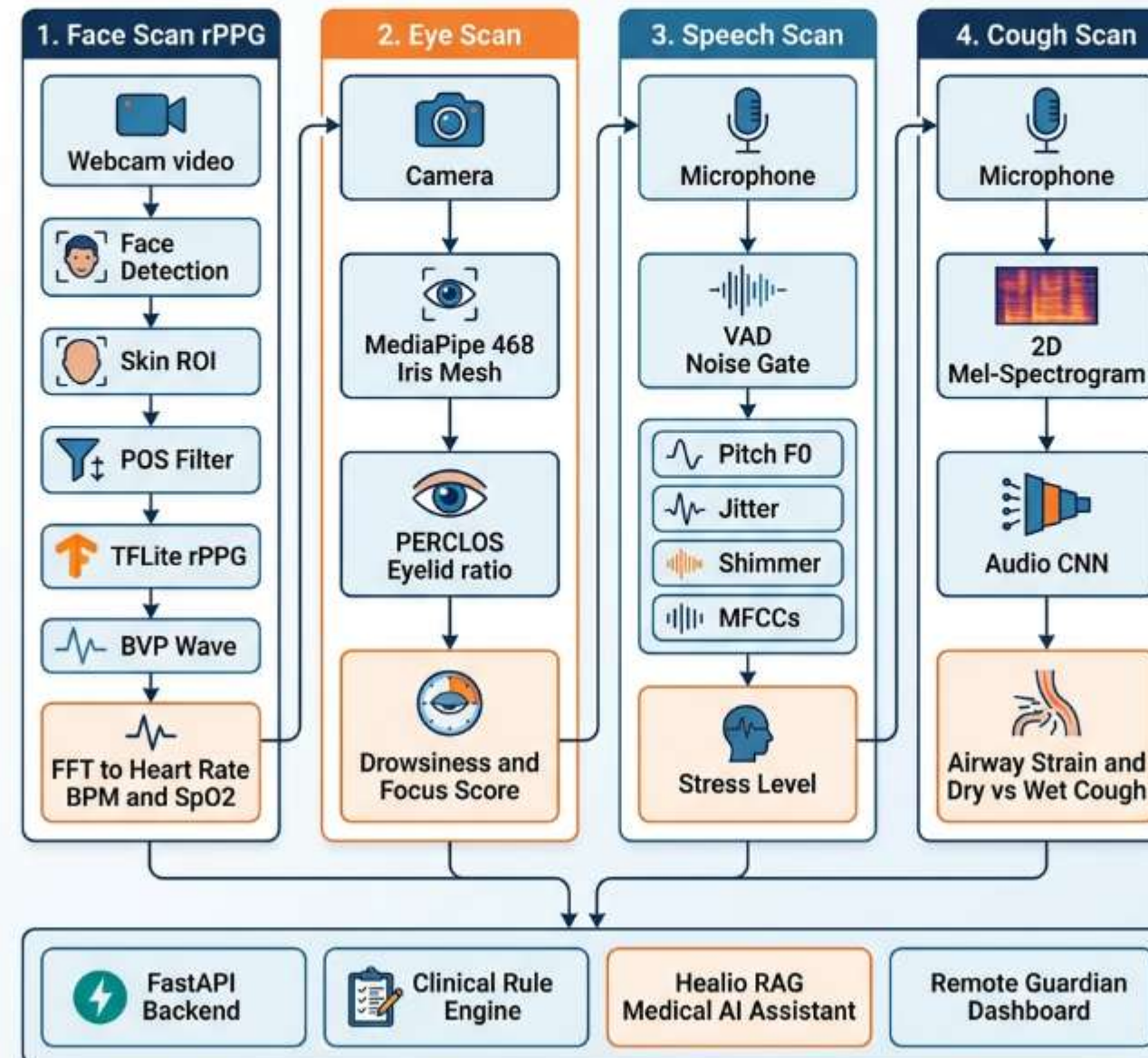
Saves trend log + sends remote guardian alerts.

Technology and System Architecture

From raw smartphone camera & mic frames to vital measurements, clinical decision scoring & Healio RAG AI.

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CareFor Multi-Modal AI System Architecture



CareFor — Multi-Modal Technology Stack

Unified stack powering 4-track AI signal ingestion, backend storage, and Healio RAG AI.

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BACKEND

- **REST API (FASTAPI)**
High performance API framework for building scalable endpoints & WebSockets.
- **PostgreSQL**
Reliable relational database for secure and efficient medical data storage.
- **Pydantic**
Data validation & settings management using Python type hints.
- **LLM (RAG)**
Retrieval Augmented Generation for context-aware medical AI responses.
- **Python**
Core programming language powering logic, APIs, & PyTorch ML models.

FRONTEND

- **Flutter**
Cross-platform UI framework for building beautiful, fast mobile apps.
- **Dart**
Programming language optimized for building modern apps with Flutter.
- **Android SDK**
Target platform for delivering CareFor app with native camera & mic access.

AI & DIAGNOSTIC MODELS

- **rPPG & Face Vitals**
Google MediaPipe + POS/CHROM Model + FFT: Skin ROI tracking & color channel filtering to extract Heart Rate (BPM), SpO2 %, & HRV.
- **Eye Fatigue Model**
MediaPipe Iris 468 Mesh + PERCLOS (ISO Standard): Calculates eyelid closure ratio $\geq 80\%$ shut & saccadic gaze velocity for Drowsiness % & Focus %.
- **Speech Acoustic Model**
Librosa VAD + Praat Pitch (F0) & Shimmer + OpenSMILE: Extracts vocal micro-trembles, pitch stability, & MFCCs to compute Stress % & Voice Stability.
- **Cough Airway AI**
STFT 2D Mel-Spectrogram + ResNet-18 Audio CNN: Classifies 1D cough sound waves into Dry vs Wet mucus patterns & Airway Strain Score %.

From Prototype to Real World

Show that the solution can be implemented, scaled and sustained.

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TECHNICAL FEASIBILITY

- **Zero Extra Hardware:** Uses standard smartphone RGB camera & mic (\$0 extra cost).
- **On-Device Processing:** MediaPipe & TFLite models run directly on mobile for fast, offline vital extraction.
- **Low-Bandwidth WebSockets:** Micro-second binary payloads transmit easily over 3G/4G/5G networks.

ECONOMIC FEASIBILITY

- **100% Free Health Check:** Eliminates costly lab tests & diagnostic gadget purchases.
- **Replaces 4 Devices:** Saves approx. ₹15,000 – ₹45,000 spent on smartwatches, oximeters, & stethoscopes.
- **Scalable Freemium Model:** Free personal screening; clinic SDK licensing & doctor dashboards.

SCALABILITY

- **Instant Global Rollout:** Zero hardware manufacturing or logistics friction; instantly downloadable app.
- **Regional Language Support:** Localized UI for Tamil, Hindi, & regional Indian languages.
- **Multi-Modal AI Growth:** Continuous model improvements with anonymized clinical feedback.

SUSTAINABILITY

- **Lightweight Cloud Stack:** Efficient FastAPI backend with PostgreSQL & vectorized RAG storage.
- **Data Privacy Standards:** Adheres to HIPAA & ABDM digital health privacy guidelines.
- **Long-Term Preventative Habit:** Empowers families with daily habit building & remote guardian alerts.

Implementation & Deployment Roadmap

A practical roadmap shows how the prototype becomes a deployable solution.

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NOW

Prototype

- **App with Fused 4-Scan AI Features:** Working Flutter app with rPPG camera vital extraction & multi-modal fusion.

- **Active Integration:** Eye fatigue, Speech acoustic, & Cough AI engines in active pipeline.

[Current capability]

NEXT

Pilot

- **Campus & Clinic Testing:** Pilot testing integrated 4-scan suite with CIT students & local clinics.

- **Ground-Truth Benchmarks:** Validating camera & mic outputs against hospital vital monitors.

[Test with users]

PHASE 2

Deployment

- **Primary Health Centers:** Deploying CareFor app across rural healthcare centers & telemedicine kiosks.

- **Remote Guardian Network:** Automatic SMS & doctor alerts on vital drops.

[Operational rollout]

FUTURE

Scale

- **Global Telehealth SDK:** Packaging CareFor engine as an embeddable SDK plugin.

- **National EHR Integration:** ABDM & HIPAA compliance for digital records.

[Expansion / integration]

SUCCESS CRITERIA

Milestone 1: Achieve $\geq 95\%$ correlation with medical pulse oximeters across diverse skin tones.

Milestone 2: Maintain sub-1-minute full 4-scan pipeline execution latency on mid-range phones.

Milestone 3: Successfully complete 500+ patient pilot trials at CIT campus clinic & rural health centers.

FROM PROBLEM TO IMPACT

